# Chapter 3 — Chemical Bonding

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

## Practice MCQs

**1. Ionic bond forms by:**

**(a)** Electron transfer **(b)** Neutron sharing **(c)** Electron sharing **(d)** Proton transfer

**2. Covalent bond forms by:**

**(a)** Metal lattice **(b)** Ion formation only **(c)** Electron transfer **(d)** Electron sharing

**3. Dative bond has shared pair donated by:**

**(a)** Nucleus **(b)** One atom **(c)** Both atoms equally **(d)** Solvent only

**4. $NH_4^+$ contains:**

**(a)** Hydrogen bond only **(b)** Only ionic bonds **(c)** Dative bond **(d)** Metallic bond

**5. Large electronegativity difference favours:**

**(a)** Non-polar covalent **(b)** London force **(c)** Metallic bond **(d)** Ionic bond

**6. $CO_2$ is non-polar because:**

**(a)** Dipoles cancel **(b)** No polar bonds **(c)** Ionic bonds **(d)** Hydrogen bonds

**7. $H_2O$ is polar because:**

**(a)** Ionic bonds **(b)** Linear shape **(c)** Bent shape **(d)** No lone pairs

**8. London forces increase with:**

**(a)** Lower electrons **(b)** Higher symmetry only **(c)** Molecular size **(d)** Smaller size

**9. Hydrogen bonding occurs with H bonded to:**

**(a)** C only **(b)** Cl only **(c)** N, O or F **(d)** Na

**10. Water has high boiling point due to:**

**(a)** Hydrogen bonding **(b)** Ionic bonds **(c)** London forces only **(d)** Metallic bonds

**11. Ice floats because:**

**(a)** Higher density **(b)** Open H-bond lattice **(c)** Ionic lattice **(d)** No bonds

**12. Shorter bond generally has:**

**(a)** Higher bond energy **(b)** Zero length **(c)** Lower bond energy **(d)** No energy

**13. C-C bond strength order:**

**(a)** Double > triple > single **(b)** All equal **(c)** Triple > double > single **(d)** Single > double > triple

**14. Sigma bond forms by:**

**(a)** Sideways overlap **(b)** Head-on overlap **(c)** No overlap **(d)** Ionic transfer

studysapiens.com

**15. Pi bond forms by:**

**(a)** s-s overlap only **(b)** Proton sharing **(c)** Sideways p overlap **(d)** Head-on overlap

**16. Double bond contains:**

**(a)** 2 sigma **(b)** 1 pi only **(c)** 1 sigma + 1 pi **(d)** 2 pi

**17. Triple bond contains:**

**(a)** 3 pi **(b)** 2 sigma + 1 pi **(c)** 1 sigma + 2 pi **(d)** 3 sigma

**18. $sp^3$ hybridization gives angle:**

**(a)** 90° **(b)** 180° **(c)** 109.5° **(d)** 120°

**19. $sp^2$ hybridization gives shape:**

**(a)** Tetrahedral **(b)** Linear **(c)** Bent **(d)** Trigonal planar

**20. sp hybridization gives shape:**

**(a)** Bent **(b)** Pyramidal **(c)** Octahedral **(d)** Linear

**21. $CH_4$ shape is:**

**(a)** Bent **(b)** Square planar **(c)** Linear **(d)** Tetrahedral

**22. $NH_3$ shape is:**

**(a)** Trigonal pyramidal **(b)** Tetrahedral **(c)** Trigonal planar **(d)** Linear

**23. $H_2O$ bond angle is about:**

**(a)** 109.5° **(b)** 120° **(c)** 104.5° **(d)** 180°

**24. $BF_3$ shape is:**

**(a)** Trigonal planar **(b)** Bent **(c)** Pyramidal **(d)** Tetrahedral

**25. $SF_6$ shape is:**

**(a)** Octahedral **(b)** Square planar **(c)** Linear **(d)** Tetrahedral

**26. VSEPR is based on repulsion of:**

**(a)** Neutrons **(b)** Electron pairs **(c)** Protons **(d)** Nuclei only

**27. Lone pairs repel:**

**(a)** Same as bond pairs **(b)** More than bond pairs **(c)** Less than bond pairs **(d)** Not at all

**28. MOT bond order formula:**

**(a)** 1/2(Nb - Na) **(b)** Nb + Na **(c)** Nb/Na **(d)** Na - Nb

**29. Bond order of $O_2$:**

**(a)** 2 **(b)** 3 **(c)** 1 **(d)** 0

studysapiens.com

**30. $O_2$ is paramagnetic due to:**

**(a)** No unpaired electrons **(b)** Ionic bonds **(c)** Two unpaired electrons **(d)** Hydrogen bonds

**31. Ionic bond forms by:**

**(a)** Electron transfer **(b)** Proton transfer **(c)** Electron sharing **(d)** Neutron sharing

**32. Covalent bond forms by:**

**(a)** Electron sharing **(b)** Electron transfer **(c)** Proton sharing **(d)** No electrons

**33. Dative bond shared pair comes from:**

**(a)** Solvent only **(b)** Both atoms equally **(c)** One atom **(d)** Nucleus

**34. Expanded octet is possible in:**

**(a)** $PCl_5$ **(b)** $NH_3$ **(c)** $CH_4$ **(d)** $H_2O$

**35. Dipole moment measures:**

**(a)** Bond/molecular polarity **(b)** Bond length **(c)** Bond energy **(d)** Atomic mass

**36. London forces arise from:**

**(a)** Hydrogen only **(b)** Instantaneous dipoles **(c)** Permanent ions **(d)** Protons

**37. Dipole-dipole forces occur between:**

**(a)** Non-polar molecules only **(b)** Ions only **(c)** Metals only **(d)** Polar molecules

**38. Hydrogen bonding is strongest when H is bonded to:**

**(a)** Cl only **(b)** C or S **(c)** N, O or F **(d)** Na or K

**39. Bond energy is energy needed to:**

**(a)** Break bonds **(b)** Form bonds **(c)** Melt solid **(d)** Ionize atom

**40. For same atoms, triple bond is:**

**(a)** Ionic **(b)** Shorter and stronger **(c)** Same as single **(d)** Longer and weaker

**41. Sigma bond overlap is:**

**(a)** Head-on **(b)** No overlap **(c)** Ionic **(d)** Sideways

**42. Pi bond overlap is:**

**(a)** Sideways **(b)** s-s only **(c)** Head-on **(d)** Proton sharing

**43. sp hybridization gives:**

**(a)** Bent shape **(b)** Tetrahedral shape **(c)** Linear shape **(d)** Trigonal planar shape

**44. $sp^2$ hybridization gives:**

**(a)** Octahedral shape **(b)** Linear shape **(c)** Tetrahedral shape **(d)** Trigonal planar shape

**45. $sp^3$ hybridization gives:**

**(a)** Linear shape **(b)** Trigonal planar shape **(c)** Bent shape **(d)** Tetrahedral shape

studysapiens.com

### 46. VSEPR says electron pairs:

**(a)** Attract strongly **(b)** Repel and spread apart **(c)** Only bond pairs matter **(d)** Do not interact

### 47. $BF_3$ shape is:

**(a)** Trigonal planar **(b)** Pyramidal **(c)** Bent **(d)** Linear

### 48. $PCl_5$ shape is:

**(a)** Linear **(b)** Octahedral **(c)** Trigonal bipyramidal **(d)** Tetrahedral

### 49. $SF_6$ shape is:

**(a)** Trigonal planar **(b)** Octahedral **(c)** Tetrahedral **(d)** Bent

### 50. Bond order equals:

**(a)** 1/2(bonding - antibonding) **(b)** antibonding only **(c)** bonding only **(d)** bonding + antibonding

studysapiens.com

## Answer Key with Brief Reasons

1. **Answer: (a) Electron transfer** — Ions attract electrostatically.
2. **Answer: (d) Electron sharing** — Shared pairs hold atoms.
3. **Answer: (b) One atom** — Coordinate bond is one-sided donation.
4. **Answer: (c) Dative bond** — N donates lone pair to $H^+$.
5. **Answer: (d) Ionic bond** — Electron transfer is favoured.
6. **Answer: (a) Dipoles cancel** — Linear shape cancels bond dipoles.
7. **Answer: (c) Bent shape** — Dipoles do not cancel.
8. **Answer: (c) Molecular size** — More electrons give stronger temporary dipoles.
9. **Answer: (c) N, O or F** — High electronegativity and lone pairs are needed.
10. **Answer: (a) Hydrogen bonding** — Strong intermolecular H bonds.
11. **Answer: (b) Open H-bond lattice** — Ice has empty spaces.
12. **Answer: (a) Higher bond energy** — Stronger attraction at shorter distance.
13. **Answer: (c) Triple > double > single** — More shared pairs strengthen bond.
14. **Answer: (b) Head-on overlap** — Axial overlap is strong.
15. **Answer: (c) Sideways p overlap** — Parallel p orbitals overlap sideways.
16. **Answer: (c) 1 sigma + 1 pi** — One axial and one sideways bond.
17. **Answer: (c) 1 sigma + 2 pi** — One sigma and two pi bonds.
18. **Answer: (c) 109.5°** — Tetrahedral arrangement.
19. **Answer: (d) Trigonal planar** — Three hybrid orbitals at 120°.
20. **Answer: (d) Linear** — Two orbitals at 180°.
21. **Answer: (d) Tetrahedral** — Four bond pairs, no lone pairs.
22. **Answer: (a) Trigonal pyramidal** — Three bond pairs and one lone pair.
23. **Answer: (c) 104.5°** — Two lone pairs compress angle.
24. **Answer: (a) Trigonal planar** — Three bond pairs, no lone pairs.
25. **Answer: (a) Octahedral** — Six bond pairs around S.
26. **Answer: (b) Electron pairs** — Pairs arrange far apart.
27. **Answer: (b) More than bond pairs** — Lone pairs are closer to central atom.
28. **Answer: (a) 1/2(Nb - Na)** — Bonding minus antibonding electrons.
29. **Answer: (a) 2** — MOT gives double bond.
30. **Answer: (c) Two unpaired electrons** — Unpaired electrons in pi* orbitals.
31. **Answer: (a) Electron transfer** — Metal to non-metal transfer.

studysapiens.com

**32. Answer: (a) Electron sharing** — Mutual sharing.

**33. Answer: (c) One atom** — Coordinate covalent bond.

**34. Answer: (a) $PCl_5$** — Period 3 can use d orbitals in book model.

**35. Answer: (a) Bond/molecular polarity** — Charge separation.

**36. Answer: (b) Instantaneous dipoles** — Temporary electron fluctuation.

**37. Answer: (d) Polar molecules** — Permanent dipoles attract.

**38. Answer: (c) N, O or F** — High electronegativity and small size.

**39. Answer: (a) Break bonds** — Gaseous bond breaking.

**40. Answer: (b) Shorter and stronger** — More shared electrons.

**41. Answer: (a) Head-on** — Along internuclear axis.

**42. Answer: (a) Sideways** — p orbitals parallel.

**43. Answer: (c) Linear shape** — 180°.

**44. Answer: (d) Trigonal planar shape** — 120°.

**45. Answer: (d) Tetrahedral shape** — 109.5°.

**46. Answer: (b) Repel and spread apart** — Minimum repulsion.

**47. Answer: (a) Trigonal planar** — Three bond pairs, no lone pair.

**48. Answer: (c) Trigonal bipyramidal** — Five bonding regions.

**49. Answer: (b) Octahedral** — Six bonding regions.

**50. Answer: (a) 1/2(bonding - antibonding)** — MOT formula.

studysapiens.com